

Product Requirements Document (PRD)

Product Name: OpenScholar

Version: 1.0

Status: Draft → MVP Planning

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1. Product Overview

1.1 Vision

To build a **global, open-access academic research platform** where students can publish, discover, and access research freely without institutional or financial barriers.

1.2 Mission

- Democratize access to academic knowledge
 - Empower students to publish research easily
 - Build a scalable academic knowledge network
 - Replace outdated, restricted research systems with a modern alternative
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1.3 Problem Statement

Current academic ecosystem problems:

- Paywalled research (Elsevier, Springer, etc.)
 - Restricted access (institutional email dependency)
 - No student-first publishing platform
 - Poor UX in existing research portals
 - Lack of visibility for student research
 - No centralized Bangladeshi research hub
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1.4 Solution Summary

OpenScholar is a:

- Free, open-access research repository
 - Student-first publishing platform
 - Modern UI/UX-driven academic system
 - Scalable SaaS platform
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2. Goals & Success Metrics

2.1 Business Goals

- Launch MVP within 4–5 months
 - Onboard 1 university (pilot)
 - Expand to 5+ universities within 1 year
 - Build strong academic network effect
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2.2 Product Goals

- Easy paper publishing (≤ 5 min process)
 - Fast discovery (< 2 sec search)
 - High readability (focus reading mode)
 - Zero paywall access
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2.3 Success Metrics (KPIs)

Metric	Target
Monthly Active Users (MAU)	10,000+
Papers Published	5,000+
Avg Session Time	5+ min
Search Response Time	<2 sec
Upload Success Rate	99%
Retention Rate	40%+

3. Target Users

3.1 Primary Users

- Undergraduate students
- Graduate students
- Research scholars

3.2 Secondary Users

- Faculty members
 - Universities
 - Public researchers
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3.3 User Personas

Student Author

- Wants to publish thesis
- Needs visibility
- Non-technical

Research Reader

- Wants free access
- Needs fast search
- Reads multiple papers

Admin

- Reviews submissions
 - Ensures quality
 - Manages platform
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4. Core Features (MVP Scope)

4.1 Authentication System

- Email/password login
 - Email verification
 - Password reset
 - Role-based access (Student/Admin)
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4.2 User Profile

- Academic profile
 - Research interests
 - Publication list
 - Profile analytics
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4.3 Research Publishing System

- Upload PDF
- Add metadata (title, abstract, keywords)
- Select university & department
- Versioning support
- Draft saving

4.4 Admin Moderation

- Approve/reject submissions
 - Feedback system
 - Content moderation
 - Duplicate detection (basic)
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4.5 Paper Viewing System

- Clean reading interface
 - Embedded PDF viewer
 - Download option
 - Citation export
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4.6 Search & Discovery

- Keyword search
 - Filters (university, year, category)
 - Fast indexing
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4.7 Dashboard

Student:

- Total uploads
- Views
- Downloads

Admin:

- Total users
- Pending approvals
- System stats

4.8 Multilingual Support

- English (default)
 - Bangla (secondary)
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5. Differentiation (USP)

OpenScholar will stand out through:

1. UI/UX First Approach

Modern, clean, readable design unlike traditional portals.

2. Focus Reading Mode

Distraction-free academic reading.

3. Discovery Modes

- Grid
- List
- Enhanced preview

4. Student-Centric Publishing

Simplified workflow vs complex journal systems.

6. User Flow (High-Level)

Publish Flow:

Register → Login → Submit Paper → Admin Review → Publish

Discover Flow:

7. System Architecture (High-Level)

Frontend:

- Next.js (App Router)
- Tailwind CSS

Backend:

- API routes (Next.js or Node.js service)

Database:

- PostgreSQL

Storage:

- Cloud storage (AWS S3 / Cloudinary)

Auth:

- NextAuth / JWT

Hosting:

- Vercel / AWS
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8. Non-Functional Requirements

Performance

- < 3s page load
- < 2s search response

Security

- Encrypted passwords
- RBAC
- Secure file uploads

Scalability

- Cloud-native
- Modular architecture

Reliability

- 99% uptime
 - Daily backups
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9. Constraints

- Initial university-only deployment
 - Limited moderation resources
 - No DOI integration (MVP)
 - No advanced plagiarism detection (MVP)
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10. Risks & Mitigation

Risk	Mitigation
Fake papers	Admin approval
Plagiarism	Manual review (MVP)
Low adoption	University onboarding
System overload	Cloud scaling

11. Roadmap

Phase 1 (MVP – 4–5 Months)

- Core platform
 - Auth
 - Publishing
 - Search
 - Admin panel
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Phase 2

- Advanced analytics
 - Better search ranking
 - Performance optimization
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Phase 3

- AI recommendations
 - Plagiarism detection
 - DOI integration
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Phase 4 (Scale)

- Multi-university onboarding
 - Global expansion
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12. Monetization (Future)

(Not for MVP but planned)

- University subscriptions

- Premium analytics tools
 - Institutional dashboards
 - API access
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13. MVP Definition (Very Important)

MVP must include ONLY:

- ✓ Auth
- ✓ Profile
- ✓ Paper upload
- ✓ Admin approval
- ✓ Search
- ✓ Read & download

Everything else = later phase.

14. Acceptance Criteria (Sample)

Paper Upload

- User can upload PDF ≤ 50MB
- Metadata required
- Status = pending

Search

- Results within 2 seconds
- Filters working correctly

Auth

- Email verification required
 - Invalid login blocked
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15. Product Philosophy

OpenScholar is built on:

- Open knowledge
- Clean design
- Accessibility
- Simplicity
- Scalability